THE CATEGORY OF PERFECT OBJECTS ON THE LIMIT IS THE COLIMIT OF THE CATEGORIES ON THE APPROXIMANTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- The comparison functor is fully faithful, which follows from the computation of Hom groups as a colimit along the descending system. invoked at proof lines 7543-7546. Reduces the statement to essential surjectivity.
- The induction principle on quasi-compact opens reduces the essential surjectivity to the affine case once the Mayer-Vietoris condition is checked with the gluing statement. invoked at proof lines 7547-7570. Organises the proof and handles the union step.
- Over an affine base a perfect complex is described concretely, and the algebra statement says that a perfect complex on the limit is the colimit of perfect complexes on the approximants. invoked at proof lines 7571-7595. Gives essential surjectivity in the affine case, which finishes the proof.

Source reference. Original lemma 08DR: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 7538-7542 to organise the proof; the Mayer-Vietoris step and the affine case are inside the counted span.

Source reference. Original lemma 09RE: sources/stacks-perfect/pinned-original-source.tex, source lines 7447–7447. Invoked at proof lines 7543-7546 to conclude full faithfulness; the remaining essential surjectivity is established inside the counted span.

Source reference. Original lemma 08DG: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 7547-7560 for the Mayer-Vietoris condition in the union step; the reduction to affines that follows is inside the counted span.

Source reference. Original lemma 08EB: sources/stacks-perfect/pinned-original-source.tex, source lines 2636–2636. Cited at proof lines 7565-7575 for the affine translation; the algebra argument that concludes the affine case is inside the counted span.

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Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 7576-7580 in the affine case; the conclusion drawn is inside the counted span.

Source reference. Original lemma 0BC7: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 7581-7590 to get essential surjectivity in the affine case; the translation to this statement is inside the counted span.

Lemma 0.2. *In Situation 0.3 the category of perfect objects of $D(\mathcal{O}_S)$ is the colimit of the categories of perfect objects of $D(\mathcal{O}_{S_i})$.*

Situation 0.3. Let $S = \lim_{i \in I} S_i$ be a limit of a directed system of schemes with affine transition morphisms $f_{i'i} : S_{i'} \to S_i$. We assume that S_i is quasi-compact and quasi-separated for all $i \in I$. We denote $f_i : S \to S_i$ the projection. We also fix an element $0 \in I$.

Proof. For every open $U_0 \subset S_0$ consider the condition P that the functor

$$\operatorname{colim}_{i \geq 0} D_{perf}(\mathcal{O}_{U_i}) \longrightarrow D_{perf}(\mathcal{O}_U)$$

is an equivalence where $_{perf}$ indicates the full subcategory of perfect objects and where $U = f_0^{-1}(U_0)$ and $U_i = f_{i0}^{-1}(U_0)$. We will prove P holds for all quasi-compact opens U_0 by the induction principle of Cohomology of Schemes, Lemma 08DR. First, we observe that we already know the functor is fully faithful by Lemma 09RE. Thus it suffices to prove essential surjectivity.

We first check condition (2) of the induction principle. Thus suppose that we have $S_0 = U_0 \cup V_0$ and that P holds for U_0, V_0, and $U_0 \cap V_0$. Let E be a perfect object of $D(\mathcal{O}_S)$. We can find $i \geq 0$ and $E_{U,i}$ perfect on U_i and $E_{V,i}$ perfect on V_i whose pullback to U and V are isomorphic to $E|_U$ and $E|_V$. Denote

$$a : E_{U,i} \to (Rf_{i,*}E)|_{U_i} \quad \text{and} \quad b : E_{V,i} \to (Rf_{i,*}E)|_{V_i}$$

the maps adjoint to the isomorphisms $Lf_i^* E_{U,i} \to E|_U$ and $Lf_i^* E_{V,i} \to E|_V$. By fully faithfulness, after increasing i, we can find an isomorphism $c : E_{U,i}|_{U_i \cap V_i} \to E_{V,i}|_{U_i \cap V_i}$ which pulls back to the identifications

$$Lf_i^* E_{U,i}|_{U \cap V} \to E|_{U \cap V} \to Lf_i^* E_{V,i}|_{U \cap V}.$$

Apply Cohomology, Lemma 08DG to get an object E_i on S_i and a map $d : E_i \to Rf_{i,*}E$ which restricts to the maps a and b over U_i and V_i. Then it is clear that E_i is perfect and that d is adjoint to an isomorphism $Lf_i^* E_i \to E$.

Finally, we check condition (1) of the induction principle, in other words, we check the lemma holds when S_0 is affine. Say $S_0 = \operatorname{Spec}(A_0)$, $S_i = \operatorname{Spec}(A_i)$, and $S = \operatorname{Spec}(A)$. Using Lemmas 06Z0 and 08EB we see that we have to show that

$$D_{perf}(A) = \operatorname{colim} D_{perf}(A_i)$$

This is clear from the fact that perfect complexes over rings are given by finite complexes of finite projective (hence finitely presented) modules. See More on Algebra, Lemma 0BC7 for details. □